DEPARTMENT OF CIVIL ENGINEERING
Indian Institute of Technology Guwahati

Subject: Civil Engineering Materials (CE 211) **Date: 22/09/2021**

Time: 9:00 - 11:00 Hrs

Answer All Questions (Briefly to the point) **Total Marks: 60**

1a) Explain about the effect of fineness of cement on properties of concrete. **[3]**

b) Write short notes on:

i) Rapid hardening Portland cement, **ii)** Sulfate resisting Portland cement, and **iii)** Portland pozzolana cement. **[3+3+3]**

c) Briefly describe about the classification of aggregates based on bulk density. **[3]**

d) Explain with reasons, which type of admixtures would you recommend for the following situations:

i) Concreting in hot weather, and **ii)** Concreting in cold weather. **[2+2]**

e) What is the effect of aggregate/cement ratio on workability of concrete mix? **[2]**

f) Explain about the effect of air entrainment on compressive strength of concrete. **[4]**

2) The oxide composition (%) of a Portland cement is as follows:
CaO = 61.3%, SiO_2 = 21.8%, Al_2O_3 = 6.2%, SO_3 = 1.9%, Alkalis = 1.1%, MgO = 2.3%, Loss on ignition = 1.7% and remaining is Fe_2O_3. Calculate the percentage of C_2S, C_3S, C_3A, and C_4AF using Bogue's equations. **[5]**

3a) What is meant by gap-graded aggregate? Show a typical particle size distribution curve for gap-graded aggregate. **[3]**

b) Explain about the mechanism of action of water reducing admixtures. **[2]**

c) Why does bleeding of concrete occur? **[2]**

d) Explain about the effect of aggregate size on compressive strength of concrete. **[3]**

e) What are the advantages of using water reducing admixtures in concrete? **[3]**

f) What are the effects of cement type on compressive strength of concrete? **[3]**

g) Explain about the mechanism of sulfate attack of concrete. **[3]**

4) Design a concrete mix by Indian Standard guidelines (IS 10262 : 2019) using the following information;
- Reinforced concrete subjected to 'very severe' exposure condition.
- Grade of concrete: M 35.
- Slump of concrete: 80 mm.
- Cement type: ordinary Portland cement (OPC) 43 grade.
- Nominal maximum size of aggregate: 20 mm.
- Type of aggregate: Crushed angular aggregate, Fine aggregate (sand): confirming to Grading Zone II.
- Specific gravity of ordinary Portland cement (OPC) 43 grade: 3.14.
- Specific gravity (SSD) of fine aggregate: 2.63, Specific gravity (SSD) of coarse aggregate: 2.68.

Calculate the quantities (by mass) of water, cement, fine aggregate (sand) and coarse aggregate for 1 m^3 of the concrete mix. **[11]**

IS 10262 : 2019

Table 1 Value of X

(*Clause* 4.2)

Sl No.	Grade of Concrete	Value of X
(1)	(2)	(3)
i)	$M10$, $M15$	5.0
ii)	$M20$, $M25$	5.5
iii)	$M30$, $M35$, $M40$, $M45$, $M50$, $M55$, $M60$	6.5
iv)	M65 and above	8.0

IS 10262 : 2019

Table 2 Assumed Standard Deviation

(*Clause* 4.2.1.3)

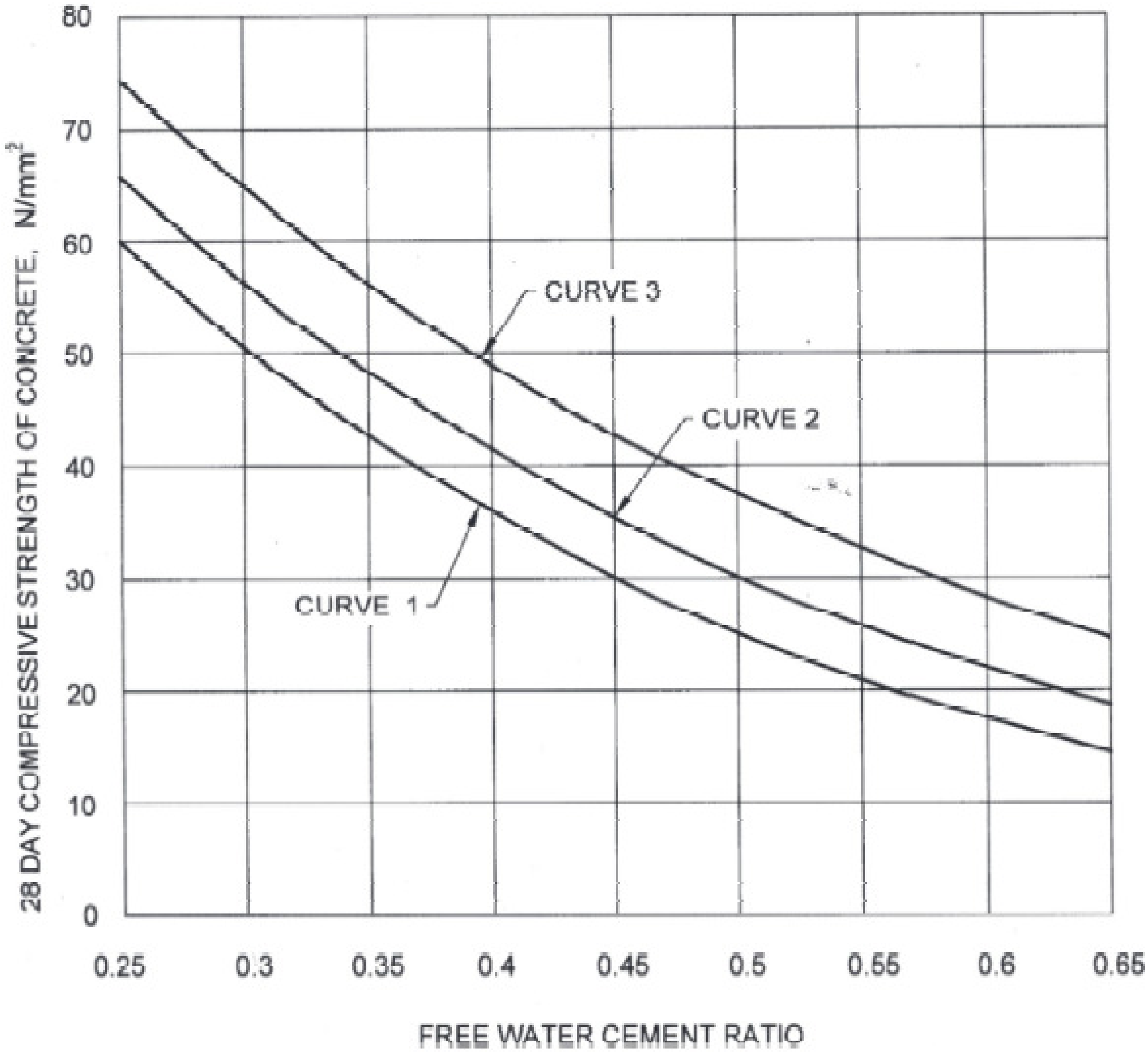
Sl No.	Grade of Concrete	Assumed Standard Deviation N/mm^2
(1)	(2)	(3)
i)	$M10$, $M15$	3.5
ii)	$M20$, $M25$	4.0
iii)	M30, M35, M40, M45, M50, M55, M60	5.0
iv)	$M65$, $M70$, $M75$, $M80$	6.0

NOTES

1 The above values correspond to good degree of site control having proper storage of cement; weigh batching of all materials; controlled addition of water; regular checking of all materials; aggregate grading and moisture content; and regular checking of workability and strength. Where there are deviations from the above, the site control shall be designated as fair and the values given in the above table shall be increased by 1 N/mm^2.

2 For grades M65 and above, the standard deviation may also be established by actual trials based on assumed proportions, before finalizing the mix.

IS 10262 : 2019



Curve 1 : for expected 28 days compressive strength of 33 and < 43 N/mm^2.
Curve 2 : for expected 28 days compressive strength of 43 and < 53 N/mm^2.
Curve 3 : for expected 28 days compressive strength of 53 N/mm^2 and above.

NOTES

1 In the absence of data on actual 28 days compressive strength of cement, the curves 1, 2 and 3 may be used for OPC 33, OPC 43 and OPC 53, respectively.

2 While using PPC/PSC, the appropriate curve as per the actual strength may be utilized. In the absence of the actual 28 days compressive strength data, curve 2 may be utilized.

FIG 1. RELATIONSHIP BETWEEN FREE WATER CEMENT RATIO AND 28 DAYS COMPRESSIVE STRENGTHS OF CONCRETE FOR CEMENTS OF VARIOUS EXPECTED 28 DAYS COMPRESSIVE STRENGTHS

IS 456 : 2000 (Reaffirmed 2021)

Table 5 Minimum Cement Content, Maximum Water-Cement Ratio and Minimum Grade of Concrete for Different Exposures with Normal Weight Aggregates of 20 mm Nominal Maximum Size

(Clauses 6.1.2, 8.2.4.1 and 9.1.2)

Sl No.	Exposure	Plain Concrete			Reinforced Concrete		
		Minimum Cement Content kg/m³	Maximum Free Water-Cement Ratio	Minimum Grade of Concrete	Minimum Cement Content kg/m³	Maximum Free Water-Cement Ratio	Minimum Grade of Concrete
1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
i)	Mild	220	0.60	–	300	0.55	M 20
iii)	Moderate	240	0.60	M 15	300	0.50	M 25
iii)	Severe	250	0.50	M 20	320	0.45	M 30
iv)	Very severe	260	0.45	M 20	340	0.45	M 35
v)	Extreme	280	0.40	M 25	360	0.40	M 40

NOTES

1 Cement content prescribed in this table is irrespective of the grades of cement and it is inclusive of additions mentioned in 5.2. The additions such as fly ash or ground granulated blast furnace slag may be taken into account in the concrete composition with respect to the cement content and water-cement ratio if the suitability is established and as long as the maximum amounts taken into account do not exceed the limit of pozzolona and slag specified in IS 1489 (Part 1) and IS 455 respectively.

2 Minimum grade for plain concrete under mild exposure condition is not specified.

IS 10262 : 2019

Table 3 Approximate Air Content

(*Clause* 5.2)

Sl No.	Nominal Maximum Size of Aggregate mm	Entrapped Air, as Percentage of Volume of Concrete
(1)	(2)	(3)
i)	10	1.5
ii)	20	1.0
iii)	40	0.8

IS 10262 : 2019

Table 4 Water Content per Cubic Metre of Concrete For Nominal Maximum Size of Aggregate

(*Clause* 5.3)

Sl No.	Nominal Maximum Size of Aggregate mm	Water Content[1)] kg
(1)	(2)	(3)
i)	10	208
ii)	20	186
iii)	40	165

[1)]Water content corresponding to saturated surface dry aggregate.

NOTES

1 These quantities of mixing water are for use in computing cement/cementitious materials content for trial batches.

2 On account of long distances over which concrete needs to be carried from batching plant/RMC plant, the concrete mix is generally designed for a higher slump initially than the slump required at the time of placing. The initial slump value shall depend on the distance of transport and loss of slump with time.

IS 10262 : 2019

Table 5 Volume of Coarse Aggregate per Unit Volume of Total Aggregate for Different Zones of Fine Aggregate for Water-Cement/Water-Cementitious Materials Ratio of 0.50

(*Clause* 5.5)

Sl No.	Nominal Maximum Size of Aggregate mm	Volume of Coarse Aggregate per Unit Volume of Total Aggregate for Different Zones of Fine Aggregate			
		Zone IV	Zone III	Zone II	Zone I
(1)	(2)	(3)	(4)	(5)	(6)
i)	10	0.54	0.52	0.50	0.48
ii)	20	0.66	0.64	0.62	0.60
iii)	40	0.73	0.72	0.71	0.69

NOTES

1 Volumes are based on aggregates in saturated surface dry condition.

2 These volumes are for crushed (angular) aggregate and suitable adjustments may be made for other shape of aggregate.

3 Suitable adjustments may also be made for fine aggregate from other than natural sources, normally, crushed sand or mixed sand may need lesser fine aggregate content. In that case, the coarse aggregate volume shall be suitably increased.

4 It is recommended that fine aggregate conforming to Grading Zone IV, as per IS 383 shall not be used in reinforced concrete unless tests have been made to ascertain the suitability of proposed mix proportions.